Answer Key

Other (130 Questions)

Q1. (A)	Q2. (D)	Q3. (A)	Q4. (C)	Q5. (C)
Q6. (A)	Q7. (C)	Q8. (C)	Q9. (C)	Q10. (D)
Q11. (C)	Q12. (D)	Q13. (C)	Q14. (A)	Q15. (A)
Q16. (B)	Q17. (A)	Q18. (C)	Q19. (A)	Q20. (C)
Q21. (D)	Q22. (A)	Q23. (B)	Q24. (D)	Q25. (B)
Q26. (B)	Q27. (C)	Q28. (C)	Q29. (A)	Q30. (B)
Q31. (C)	Q32. (A)	Q33. (A)	Q34. (B)	Q35. (B)
Q36. (B)	Q37. (B)	Q38. (C)	Q39. (A)	Q40. (B)
Q41. (B)	Q42. (D)	Q43. (A)	Q44. (D)	Q45. (B)
Q46. (D)	Q47. (C)	Q48. (B)	Q49. (C)	Q50. (D)
Q51. (D)	Q52. (A)	Q53. (C)	Q54. (D)	Q55. (D)
Q56. (B)	Q57. (D)	Q58. (A)	Q59. (B)	Q60. (B)
Q61. (C)	Q62. (A)	Q63. (A)	Q64. (B)	Q65. (A)
Q66. (D)	Q67. (A)	Q68. (B)	Q69. (A)	Q70. (C)
Q71. (C)	Q72. (C)	Q73. (B)	Q74. (D)	Q75. (B)
Q76. (C)	Q77. (A)	Q78. (C)	Q79. (B)	Q80. (A)
Q81. (B)	Q82. (A)	Q83. (A)	Q84. (C)	Q85. (D)
Q86. (B)	Q87. (B)	Q88. (C)	Q89. (C)	Q90. (A)
Q91. (C)	Q92. (C)	Q93. (A)	Q94. (D)	Q95. (B)
Q96. (A)	Q97. (C)	Q98. (C)	Q99. (A)	Q100.(A)
Q101.(C)	Q102.(D)	Q103.(A)	Q104.(A)	Q105.(C)

Q106.(A)	**Q107.(D)**	**Q108.(C)**	**Q109.(C)**	**Q110.(C)**
Q111.(C)	**Q112.(D)**	**Q113.(B)**	**Q114.(A)**	**Q115.(A)**
Q116.(B)	**Q117.(B)**	**Q118.(D)**	**Q119.(C)**	**Q120.(A)**
Q121.(D)	**Q122.(B)**	**Q123.(A)**	**Q124.(D)**	**Q125.(B)**
Q126.(B)	**Q127.(C)**	**Q128.(C)**	**Q129.(B)**	**Q130.(D)**

Solutions

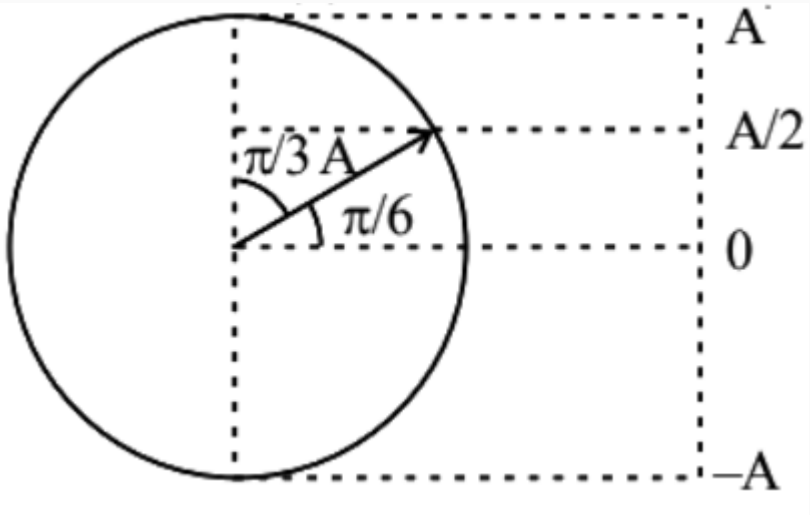
Q1. Solution

Correct Answer: (A)

As. F(air)=F(medium) $\Rightarrow \frac{1}{4\pi\epsilon_0}\frac{q_1q_2}{d_{air}^2} = \frac{1}{K}\frac{1}{4\pi\epsilon_0}\frac{q_1q_2}{d^2} \Rightarrow d_{air} = d\sqrt{K}$

Q2. Solution

Correct Answer: (D)



Let time from 0 to $A/2$ is t_1 and from $A/2$ to A is t_2 From the standard equation of SHM, $\Rightarrow \frac{A}{2} = A\sin(\omega t_1)$ $\Rightarrow \omega t_1 = \sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$ ....(i) Using $x = A_0 \sin\omega t$ again $A = A\sin\omega(t_1+t_2)$ $\omega(t_1+t_2) = \sin^{-1}(1) = \frac{\pi}{2}$ Using (i) $\omega t_2 = \frac{\pi}{2} - \frac{\pi}{6} = \frac{\pi}{3}$ .....(ii) Dividing equation (i) by (ii) we get $\frac{t_1}{t_2} = \frac{1}{2}$

$\Rightarrow \quad t_2 = 2t_1 = 2\times 2 = 4\text{sec}$

Q3. Solution

Correct Answer: (A)

Let Initial length $= l_1$ Final length $= l_2$ Initial area $= A_1$ Final area $= A_1$ $\therefore$ Volume remains same $\therefore A_1l_1 = A_2l_2 \Rightarrow A_1l_1 = A_2\frac{l_1}{4} \Rightarrow 4A_1 = A_2$ Initial resistance, $R_1 = \frac{\rho l_1}{A_1} = 160\Omega$ (given) Final resistance, $R_2 = \frac{\rho l_2}{A_2}$ $\therefore \frac{R_2}{R_1} = \frac{l_2 A_1}{A_2 l_1} = \frac{l_1}{4}\frac{A_1}{4A_1 l_1} \Rightarrow R_2 = \frac{1}{16}R_1 = \frac{1}{16}\times 160 = 10\Omega$

Q4. Solution

Correct Answer: (C)

Heat, $Q = mL$ where, $L =$ latent heat $\therefore L = \frac{Q}{m} = \frac{ML^2T^{-2}}{M} = M^0L^2T^{-2}$

Q5. Solution

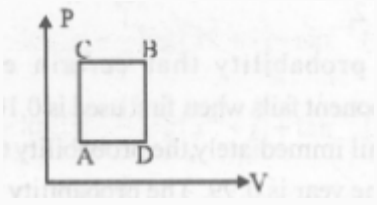
Correct Answer: (C)

Electric potential at a point P due to a point charge, $\left(\because K = 9\times 10^9\right)$ $V_P = \frac{KQ}{r} \Rightarrow \quad 50 = \frac{9\times 10^9\times 5\times 10^{-9}}{r}$

$\Rightarrow \quad r = \frac{45}{50} = \frac{9}{10} = 0.9\text{ m} = 90\text{ cm}$

Q6. Solution

Correct Answer: (A)



$\triangle$U remains same for both paths ACB and ADB $\Delta Q_{ACB} = \Delta W_{ACB} + \Delta U_{ACB}$ $\Rightarrow 60 \text{ J} = 30 \text{ J} + \Delta \mathrm{U}_{\mathrm{ACB}} \Rightarrow \mathrm{U}_{\mathrm{ACB}} = 30 \text{ J}$ As change in internal energy depends only on initial and final point $\therefore \Delta \mathrm{U}_{\mathrm{ADB}} = \Delta \mathrm{U}_{\mathrm{ACB}} = 30 \text{ J}$ $\Delta \mathrm{Q}_{\mathrm{ADB}} = \Delta \mathrm{U}_{\mathrm{ADB}} + \Delta \mathrm{W}_{\mathrm{ADB}} = 10 \text{ J} + 30 \text{ J} = 40 \text{ J}$

Q7. Solution

Correct Answer: (C)

Displacement, $x = t^3 - 6t^2 + 20t + 15$ $\therefore$ Velocity, $v = \frac{\mathrm{dx}}{\mathrm{dt}} = 3\mathrm{t}^2 - 12\mathrm{t} + 20$ $\therefore$ Acceleration, $a = \frac{dv}{dt} = 6t - 12$ When $a = 0 \Rightarrow 6\mathrm{t} - 12 = 0 \Rightarrow t = 2$ s At t = 2 s, $v = 3(2)^2 - 12(2) + 20 = 8 \text{ m/s}$

Q8. Solution

Correct Answer: (C)

Magnetic field due to straight wire, $B = \frac{\mu_0 I}{2\pi r}$ Magnetic field at $\frac{a}{2}$ is, $\mathrm{B}_{\mathrm{a}/2} = \frac{\mu_0 \mathrm{I}}{2\pi(\mathrm{a}/2)}$ $\therefore \frac{\mathrm{B}_{\mathrm{a}/2}}{\mathrm{B}_{2\mathrm{a}}} = \frac{1}{1} = 1:1$

Q9. Solution

Correct Answer: (C)

Impedance in LCR circuit $Z = \sqrt{(X_L - X_C)^2 + R^2} \quad \because X_L = X_C = R \therefore Z = R$,

Q10. Solution

Correct Answer: (D)

$v = \frac{C}{\sqrt{\mu_{\mathrm{r}} \epsilon_{\mathrm{r}}}} \Rightarrow 1.5 \times 10^8 = \frac{3 \times 10^8}{\sqrt{2 \times \epsilon_{\mathrm{r}}}} \Rightarrow 2\epsilon_{\mathrm{T}} = 4 \Rightarrow \epsilon_{\mathrm{T}} = 2$

Q11. Solution

Correct Answer: (C)

Range of projectile $R = \frac{v^2 \sin 2\theta}{g} (\because R \propto \sin(2\theta))$ $\frac{R_1}{R_2} = \frac{\sin(2\theta_1)}{\sin(2\theta_2)} = \frac{\sin(2 \times 15)}{\sin(2 \times 45)} = \frac{\sin 30^\circ}{\sin 90^\circ}$ $\Rightarrow \frac{50}{\mathrm{R}_2} = \frac{1}{2} \Rightarrow \mathrm{R}_2 = 100 \text{ m}$ ~

Q12. Solution

Correct Answer: (D)

The rate of mutual inductance is given by $\mathrm{M} = \mu_0 \mathrm{n}_1 \mathrm{n}_2 \pi \mathrm{m}_1^2$ ....(i) The rate of self inductance is given by $\mathrm{L} = \mu_0 \mathrm{n}_1^2 \pi \mathrm{r}_1^2$ .....(ii) Dividing (i) by (ii) $\Rightarrow \frac{\mathrm{M}}{\mathrm{L}} = \frac{\mathrm{n}_2}{\mathrm{n}_1}$

Q13. Solution

Correct Answer: (C)

Given lights are of same wavelength and stopping potential is independent on intensity. Hence stopping potential will remain same. Intensity $\mathrm{I}_2 > \mathrm{I}_1$, hence saturation current corresponding to I_2 will be greater than that corresponding to I_1. ~

Q14. Solution

Correct Answer: (A)

Acceleration is given as: $a = \left(\frac{m_2 - m_1}{m_1 + m_2}\right) g \Rightarrow \frac{\mathrm{g}}{\sqrt{2}} = \left(\frac{\mathrm{m}_2 - \mathrm{m}_1}{\mathrm{m}_1 + \mathrm{m}_2}\right) \mathrm{g} \Rightarrow \sqrt{2}\left(m_2 - m_1\right) = m_1 + m_2$
$\Rightarrow \frac{\mathrm{m}_1}{\mathrm{m}_2} = \left(\frac{\sqrt{2}-1}{\sqrt{2}+1}\right)$,

Q15. Solution

Correct Answer: (A)

Wavelength of H -atom is $\frac{1}{\lambda} = \mathrm{Rz}^2\left(\frac{1}{\mathrm{n}_1^2} - \frac{1}{\mathrm{n}_2^2}\right)$ Shortest wavelength for Balmer series: $\frac{1}{\lambda_B} = \mathrm{Rz}^2\left(\frac{1}{2^2} - \frac{1}{\infty}\right)$ Shortest wavelength for Lyman series: $\frac{1}{\lambda_{\mathrm{L}}} = \mathrm{Rz}^2\left(\frac{1}{1^2} - \frac{1}{\infty}\right)$ ....(ii) Dividing eq. (i) and (ii), $\lambda_{\mathrm{A}} : \lambda_1 = 4 : 1$.

Q16. Solution

Correct Answer: (B)

Mass of block, $m = 1$ kg Force of parallel inclined surface, $\mathrm{F} = 10$ N Work done against frictional force $= \mu_2$ N $\times 10 = \mu \mathrm{Mg} \cos 60 \times 10 = 0.1 \times 5 \times 10 = 5$ J ^

Q17. Solution

Correct Answer: (A)

$\because \mathrm{f}_0 + \mathrm{f}_{\mathrm{e}} = 30$ And magnification, $\mathrm{m} = \frac{\mathrm{f}_0}{\mathrm{f}_e}\ 2 = \frac{\mathrm{f}_0}{\mathrm{f}_{\mathrm{e}}} \Rightarrow \mathrm{f}_0 = 2\mathrm{f}_{\mathrm{e}} \Rightarrow \mathrm{f}_0 + \frac{\mathrm{f}_0}{2} = 30 \therefore \mathrm{f}_0 = 20cm$ ^

Q18. Solution

Correct Answer: (C)

${}_6\mathrm{C}^{13}+$ Energy $\rightarrow {}_6\mathrm{C}^{12} + {}_0\mathrm{n}^1$ Mass defect, $\Delta \mathrm{m} = (12.000000 + 1.008665) - 13.003354 = -0.00531\mathrm{u} \therefore$ Energyrequired $= \Delta \mathrm{m} \times 931.5 = 0.00531 \times 931.5$ MeV $= 4.95$MeV ^

Q19. Solution

Correct Answer: (A)

Angular momentum, $L = mvH = mu \cos 30H = mu \cos 30^\circ \times \frac{u^2 \sin^2 \theta}{2g}\left[\because H = \frac{u^2 \sin^2 \theta}{2g}\right]$
$= \frac{mu^3}{2g} \times \frac{\sqrt{3}}{2} \times \left(\frac{1}{2}\right)^2 = \frac{\sqrt{3} mu^3}{16g}$!

Q20. Solution

Correct Answer: (C)

Energy is, $\mathrm{E}=\frac{\mathrm{hc}}{\lambda}$ $\mathrm{E}=\frac{1242\ \mathrm{nm-eV}}{\lambda} \Rightarrow \lambda=\frac{1242}{6}=207\ \mathrm{nm}$,

Q21. Solution

Correct Answer: (D)

Basically 1 year is equal to time period of earth revolution around sun. So, $T_i=1$ year Now, $T^2 \propto R^3$ $\therefore\left(\frac{T_2}{T_1}\right)^2=\left(\frac{R_2}{R_1}\right)^3 \Rightarrow T_2=\left(\frac{R_2}{R_1}\right)^{3/2} T_1=\left(\frac{3R}{R}\right)^{3/2} \times 1=3\sqrt{3}$ years.

Q22. Solution

Correct Answer: (A)

At STP Temperature, $T=273\ \mathrm{K}$ Molecular mass of oxygen, $\mathrm{M}=32 \times 10^{-3}$ kg Speed of sound is given by $\mathrm{v}=\sqrt{\frac{\gamma \mathrm{RT}}{\mathrm{M}}}=\sqrt{\frac{1.4 \times 8.3 \times 273}{32 \times 10^{-3}}}=314.8541 \simeq 315\ \mathrm{m/s}$

Q23. Solution

Correct Answer: (B)

Output $\mathrm{Y}=\overline{\overline{\mathrm{A}} \cdot \overline{\mathrm{B}}}=\overline{\overline{\mathrm{A}+\mathrm{B}}}$ (ByDe-Morgan Law) $\therefore Y=\mathrm{A}+\mathrm{B}$ This Boolean expression represents OR gate.

Q24. Solution

Correct Answer: (D)

RMS speed, $\mathrm{V}_{\mathrm{rms}}=\sqrt{\frac{3 \mathrm{RT}}{\mathrm{M}}} \Rightarrow \quad \mathrm{V}_{\mathrm{mm}} \propto \sqrt{\mathrm{T}}$ Here, $\mathrm{T}_{\text{initiol}}=200\ \mathrm{K}$ $\mathrm{T}_{\text{fing1}}=800\ \mathrm{K}$ Initial RMS speed $=\mathrm{v}_0$ $\therefore \frac{v_0}{v_{\mathrm{rms}}}=\sqrt{\frac{200}{800}} \Rightarrow v_{\mathrm{ms}}=2 v_0$

Q25. Solution

Correct Answer: (B)

Young's modulus, $\mathrm{Y}=\frac{\text{Stress}}{\text{Strain}}$ If the temperature increases, strain also increases. Hence young's modulus decreases.

Q26. Solution

Correct Answer: (B)

Let us complete the reactions: (i) $\mathrm{Fe}^{2+}+\mathrm{H}_2\mathrm{O}_2^- \rightarrow \mathrm{Fe}^{3+}+\mathrm{H}_2\mathrm{O}^{2-}$ (ii) $\mathrm{Mn}^{7+}\mathrm{O}_4^-+\mathrm{H}^++\mathrm{H}_2\mathrm{O}_2^- \rightarrow \mathrm{Mn}^{2+}+\mathrm{O}_2^0+\mathrm{H}_2\mathrm{O}$ (iii) $\mathrm{I}_2{}^{\mathrm{O}}+\mathrm{H}_2\mathrm{O}_2^-+\mathrm{OH}^- \rightarrow \mathrm{I}^-+\mathrm{O}_2{}^0+\mathrm{H}_2\mathrm{O}$ (iv) $\mathrm{Mn}^{2+}+\mathrm{H}_2\mathrm{O}_2^- \rightarrow \mathrm{Mn}^{4+}+\mathrm{OH}^{2-}$ In reactions, Fe^{2+} of (i) and Mn^{2+} of (iv) get oxidised to Fe^{3+} and $\mathrm{Mn}n^{4+}$ respectively by $\mathrm{H}_2\mathrm{O}_2$, which itself is reduced to $\mathrm{H}_2\mathrm{O}$.

Q27. Solution

Correct Answer: (C)

The colloidal particles have a tendency to preferentially adsorb a particular type of ions form the solution. This preferential adsorption of a particular type of ions imparts a particular type of charge to colloidal narticles.

Q28. Solution

Correct Answer: (C)

We know, $r_1 = k[A]^{\mathrm{n}}$ ....(i) If conc. of A is increased four times rate of reaction on doubles. $2r_1 = k[4A]^{\mathrm{n}}$ ....(ii) Divide Eq. (ii) by (i). $\frac{2r_i}{r_1} = \frac{k(4A)^n}{k(A)^n} \Rightarrow \quad 2 = (4)^{\mathrm{n}} \Rightarrow 2\mathrm{n} = (2)^{2\mathrm{n}} \Rightarrow n = 0.5$

Q29. Solution

Correct Answer: (A)

In reactivity series, Cu lies below Hydrogen and it is least electropositive among the given metals.

Q30. Solution

Correct Answer: (B)

Given 1 Main scale division = m n$MSD = (\mathrm{n}+1)\mathrm{VSD} \Rightarrow \quad 1\mathrm{VSD} = \frac{\mathrm{n}}{\mathrm{n}+1}\mathrm{MSD}$ Least count of vernier caliper $= 1\mathrm{MSD} - 1\mathrm{VSD} \Rightarrow \quad L.C = m - m\left(\frac{n}{n+1}\right) = \left(1 - \frac{n}{n+1}\right)m$
$= m\left(\frac{n+1-n}{n+1}\right) = \left(\frac{1}{n+1}\right)m \Rightarrow L \cdot C = \left(\frac{m}{n+1}\right)$

Q31. Solution

Correct Answer: (C)

According to question, pressure inside, 1st soap bubble, $\Delta P_1 = P_1 - P_0 = 0.01 = \frac{4T}{R_1}$ .....(i) And $\Delta P_2 = P_2 - P_0 = 0.02 = \frac{4T}{R_2}$ .....(ii) Dividing, equation (ii) by (i), $\frac{1}{2} = \frac{R_2}{R_1} \Rightarrow R_1 = 2R_2$ Volume $V = \frac{4}{3}\pi R^3 \Rightarrow \frac{V_1}{V_2} = \frac{R_1^3}{R_2^3} = \frac{8R_2^3}{R_2^3} = \frac{8}{1}$

Q32. Solution

Correct Answer: (A)

Condition for minima is $\mathrm{d}\sin\theta = \mathrm{n}\lambda$ For Ist minimad $\sin\theta = \lambda \sin\theta = \frac{\lambda}{a} = \frac{2}{4} = \frac{1}{2} \quad \therefore \theta = 30^\circ$ Angular spread $= 2\theta = 60^\circ$

Q33. Solution

Correct Answer: (A)

Position of COM of a mass - system is given as, $\vec{r}_{\mathrm{com}} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2}{m_1+m_2} = \frac{1(\hat{i}+2\hat{j}+\hat{k})+3(-3\hat{i}-2\hat{j}+\hat{k}}{1+3} = -2\hat{i} - \hat{j} + \hat{k}$
$\vec{r}_{cm} = |-2\hat{i} - \hat{j} + \hat{k}| = \sqrt{(2)^2 + (1)^2 + (1)^2} = \sqrt{6}$ Only option (1) magnitude is $\sqrt{1^2 + (-2)^2 + 1^2} = \sqrt{6}$ So option (1) is correct.

Q34. Solution
Correct Answer: (B)

We have $\Delta V = V_0 \gamma \Delta T$ $\Delta V = a^3, (3\alpha)\Delta T$ Now, $6a^2 = 24 \quad [\because$ Total surface area of cube $= 6a^2]$
$\Rightarrow a^2 = 4 \Rightarrow a = 2$ So, $\Delta V = 2^3 \left(3 \times 5 \times 10^{-4}\right) \times 10 = 1200 \times 10^{-4}\ \mathrm{m}^3$
$= 1200 \times 10^2\ \mathrm{cm}^3 = 1.2 \times 10^5\ \mathrm{cm}^3$

Q35. Solution
Correct Answer: (B)

$1\mathrm{M} = 1$ mole of solute in 1 L solution.

Q36. Solution
Correct Answer: (B)

Smaller the size of anion, smaller is the covalent character and higher will be its melting point. Hence, the order of melting point of lithium salts is $\mathrm{LiF} > \mathrm{LiCl} > \mathrm{LIBr}$.

Q37. Solution
Correct Answer: (B)

During the extraction of copper, iron is present in the ore as impurity (FeS). The ore together with a little coke and silica is smelted; FeS present as impurity in the ore is oxidized to iron oxide, which then reacts with silica to form fusible ferrous silicate which is removed as slag. $2\mathrm{FeS} + 3\mathrm{O}_2 \rightarrow 2\mathrm{FeO} + 2\mathrm{SO}_2 \uparrow$
$\mathrm{FeO} + \mathrm{SiO}_2 \rightarrow \underset{\text{(Slag)}}{\mathrm{FeSiO}_3}$

Q38. Solution
Correct Answer: (C)

Saline hydrides are not volatile as the strong ionic bonds keep the constituent ions together. Thus, statement II is incorrect. Electron-precise hydrides like $\mathrm{CH}_4, \mathrm{SiH}_4$, act as pH - neutral species. Thus, statement 3 is incorrect.

Q39. Solution
Correct Answer: (A)

$\mathrm{H} = \mathrm{U} + \mathrm{PV}$ (By definition) $\Delta\mathrm{H} = \Delta\mathrm{U} + \Delta(\mathrm{PV})$ at constant pressure $\Delta H = \Delta U + P\Delta V$

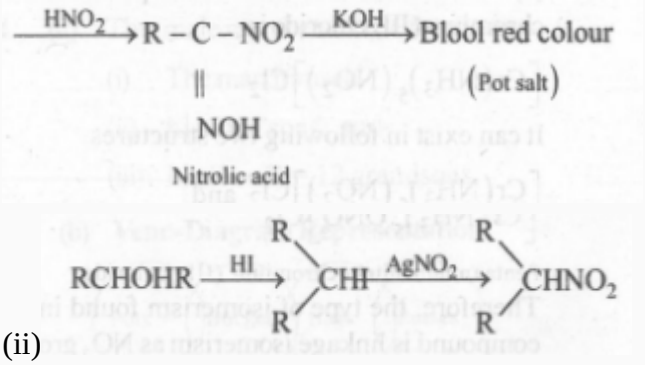
Q40. Solution
Correct Answer: (B)

The increasing order of field strength of ligands (according to spectrochemical series)
$\mathrm{S}^{2-} < \mathrm{C}_2\mathrm{O}_4^{2-} < \mathrm{NH}_3 < \mathrm{en} < \mathrm{CO}$

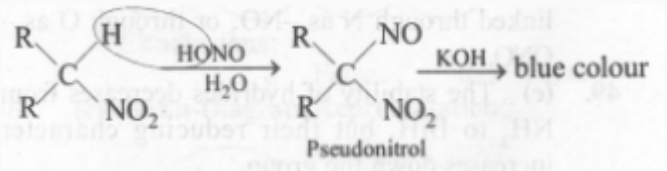
Q41. Solution

Correct Answer: (B)

Victor Meyer's test: The various steps involved are (i)$\mathrm{RCH_2OH} \xrightarrow{\mathrm{HI}} \mathrm{RCH_2I} \xrightarrow{\mathrm{AgNO_2}} \mathrm{RCH_2NO_2}$



(ii)



(iii) $\mathrm{R_3COH} \xrightarrow{\mathrm{HI}} \mathrm{R_3CI} \xrightarrow{\mathrm{AgNO_2}} \mathrm{R_3CNO_2} - \xrightarrow{\mathrm{HNO_2}}$ No reaction $\xrightarrow{\mathrm{KOH}}$ Colourless

Q42. Solution

Correct Answer: (D)

B group vitamins and vitamin C are soluble in water except vitamin $\mathrm{B_{12}}$ all are excreted easily.

Q43. Solution

Correct Answer: (A)

$\log\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$ or, $\log\left(\frac{2k}{k}\right) = \frac{E_a}{2.303\times 8.3} \times \frac{[600-300]}{300\times 600}$ $E_a \approx 3.45\ \mathrm{kJ/mol}$

Q44. Solution

Correct Answer: (D)

$\mathrm{XeF_2}$ has three lone pairs and two bond pairs. The three lone pairs are arranged in the equatorial positions because of which they face maximum repulsion from each other.

Q45. Solution

Correct Answer: (B)

Ferocene is $\mathrm{Fe}\left(\eta^5 - \mathrm{C_5H_5}\right)_2$ Perxenate ion is $[\mathrm{XeO_6}]^4$ which is octahedral.

Q46. Solution

Correct Answer: (D)

Sodium stearate is used in soap and sodium lauryl sulphate $CH_3(CH_2)_{10}CH_2 - OSO_3^{\ominus}Na^{\oplus}$ is an anionic detergent

Q47. Solution

Correct Answer: (C)

$A + B^{2+} \rightleftharpoons A^{2+} + B$ $G^{\circ} = -RT \ln K_c = -8.314 \times 298 \times 2.303 \times \log 10^{12} = -68.47\ kJ/$ mol

Q48. Solution

Correct Answer: (B)

The chemical formula of nitropenta ammine chromium(III) chloride is: $[Cr(NH_3)_5\ (NO_2)]Cl_2$ It can exist in following two structures $[Cr(NH_3)_5\ (NO_2)]Cl_2$ and Pentaamminenitrochromium (III) chloride $[Cr(NH_3)_5ONO]Cl_2$ Pentaamminenitritochromium (III) chloride Therefore, the type of isomerism found in this compound is linkage isomerism as NO_2 group is linked through N as $-NO_2$ or through O as ONO.

Q49. Solution

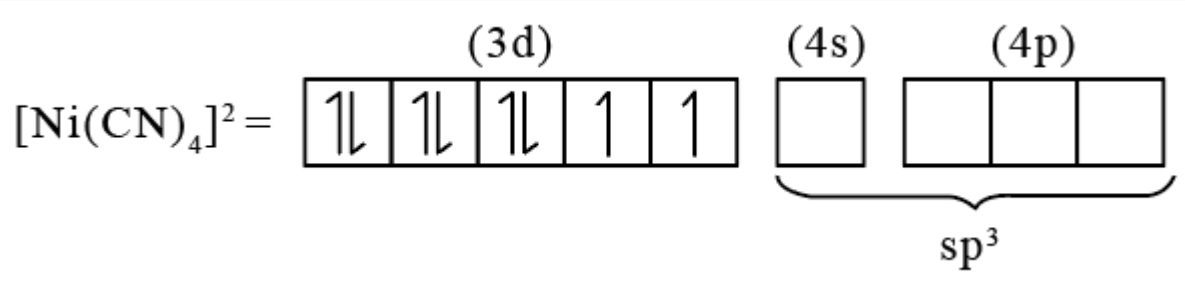
Correct Answer: (C)

The stability of hydrides decreases from NH_3 to BiH_3 but their reducing character increases down the group.

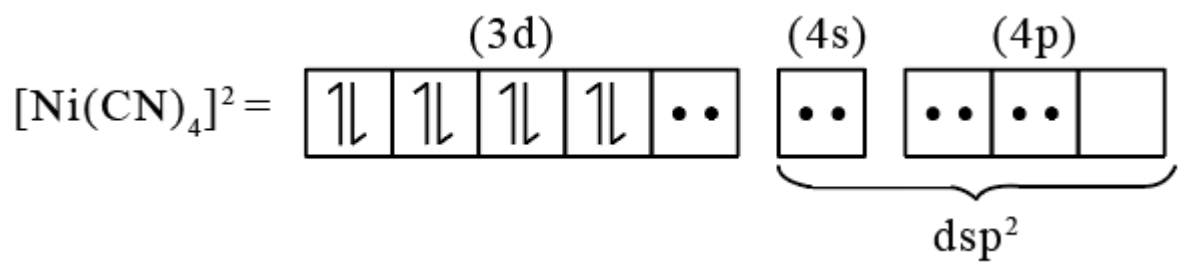
Q50. Solution

Correct Answer: (D)

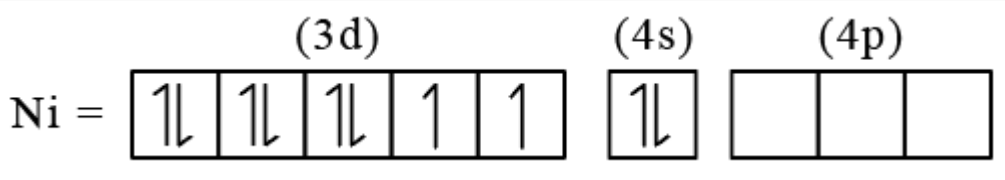
(a) Ni in $[\mathrm{Ni(Cl)_4}]^{2-}$ exist as Ni^{2+} ion. Cl is a weak field ligand (high spin). It will not cause pairing of electrons. Hence, configuration of $[\mathrm{Ni(Cl)_4}]^{2-}$ is



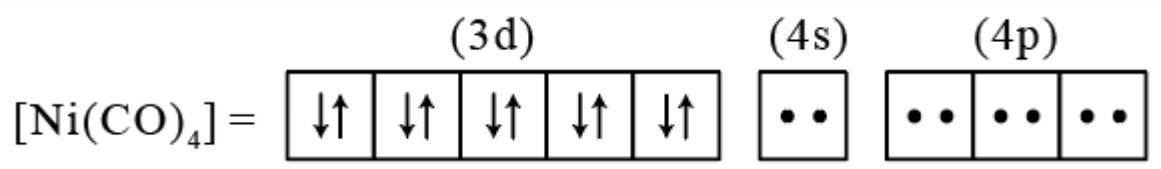
Tetrahedral with two unpaired electrons (i.e., paramagnetic) and has sp^3 hybridisation. (b) In $[\mathrm{Co(C_2O_4)_3}]^{3-}$, $(\mathrm{C_2O_4})_3^{2-}$ is a bidentate ligand thus, give octahedral structure. (c) In $[\mathrm{Ni(CN)_4}]^{2-}$, Ni exist as Ni^{2+} ion. CN^- is a strong field ligand (low spin). Causes pairing of electrons of $3d$ orbital.



Hence, the structure of is square planar and it is diamagnetic. (d) In $[\mathrm{Ni(CO)_4}]$, Ni has zero oxidation state. i.e.,



CO is a strong field ligand causes rearrangement and pairing of electrons of $3d$ and $4s$ orbital.



Structure of $\mathrm{Ni\,(CO_4)}$ is tetrahedral with sp^3 hybridisation and it is diamagnetic. Hence, (4) is the correct answer.

Q51. Solution

Correct Answer: (D)

The given reaction is for the industrial preparation methanol from a mixture of carbon monoxide and hydrogen . In the reaction the gaseous mixture is subjected to 200 atmosphere and then passed over heated catalyst mixture of ZnO and $\mathrm{Cr_2O_3}$ kept at $573-673$ K. This reaction results the formation of methanol vopours which are then condensed to liquid state.

Q52. Solution

Correct Answer: (A)

cis-polyisoprene is natural rubber.

Q53. Solution

Correct Answer: (C)

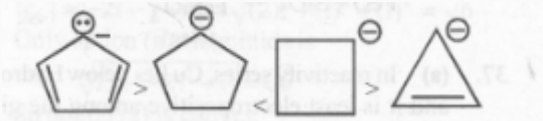
$$\frac{\text{Mass of metal}_1\text{ deposited}}{\text{Mass of metal}_2\text{ deposited}} = \frac{\text{Eq.wt.of metal}_1}{\text{Eq.wt.of metal}_2}$$

The equivalent weight of silver is highest among the given options.

Q54. Solution

Correct Answer: (D)

As we know compound (4) is aromatic and the compound (1) is anti-aromatic. Hence compound (4) is most stable and compound (1) is least stable among these in compound (2) and (3) carbon atom having charge is sp^3 hybridised. On the basis of angle strain theory compound (3) is more stable than compound (2).



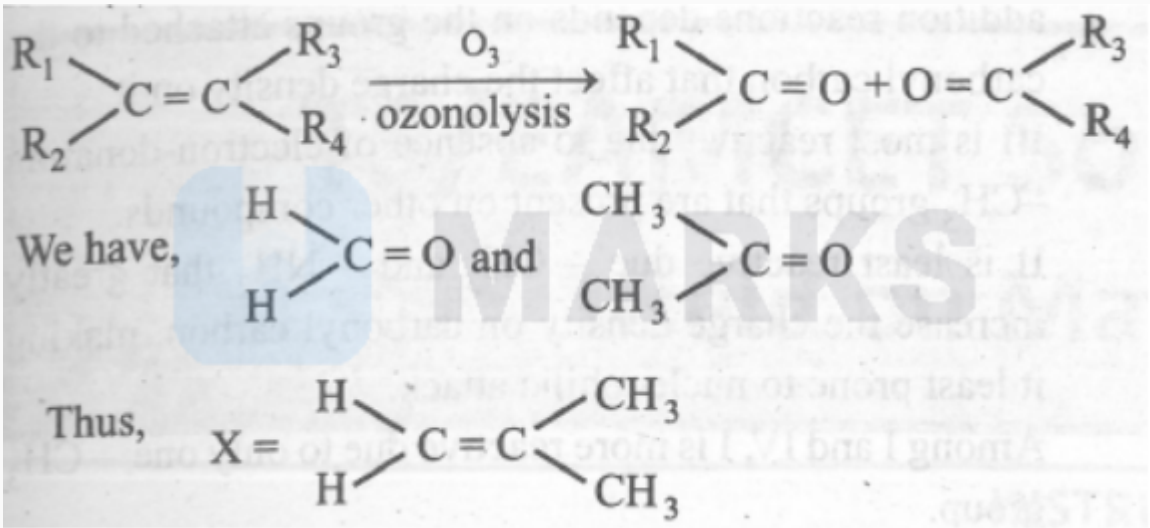
Q55. Solution

Correct Answer: (D)

Grignard reagent $= \mathrm{R}-\overset{\delta+}{\mathrm{Mg}}\overset{\delta-}{\mathrm{X}}$

Q56. Solution

Correct Answer: (B)



Q57. Solution

Correct Answer: (D)

The reaction $\mathrm{P_4 + 8SOCl_2 \rightarrow 4PCl_3 + 2\ S_2Cl_2 + 4SO_2}$ involves change of oxidation state of P from 0 to +3 and that of S from +4 to +2 . Thus, it is a redox reaction but not a disproportionation reaction.

Q58. Solution
Correct Answer: (A)

no. of moles of CH_4 $(n_1) = \frac{4}{16} = 0.25$ mol no. of moles of CO_2 $(n_2) = \frac{4.4}{44} = 0.1$ mol Total no. of moles $(n_T) = n_1 + n_2 = 0.35$ mol $\therefore \quad P = \frac{0.35 \times 0.082 \times 300}{1} = 8.6$ atm

Q59. Solution
Correct Answer: (B)

Le-Chatelier principle is not applicable to pure solids and liquids because they experience negligible change in concentration during chemical equilibrium.

Q60. Solution
Correct Answer: (B)

$E_n = -13.6\left[\frac{z^2}{n^2}\right]eV \Rightarrow \quad E_1 = -13.6\left[\frac{3^2}{1^2}\right]\text{eV} = -122.4\text{eV}$
$= -122.4 \times 1.602 \times 10^{-19}$ J $= -1.962 \times 10^{-17}$ J

Q61. Solution
Correct Answer: (C)

DBAC - Let's understand the reason for chosen option: - Option D: This sentence talks about the great sage who lives in a small village - Option B: This sentence talks about the occupation of the sage which attracted a thousand visitors to him. - Option A: This sentence talks about his busy schedule that he had no time to eat. - Option C: This sentence talks about people who did not care about him whether he had eaten something or had all the basic things for survival. - The grammatically and contextually correct sequence is 'DBAC'. The correct sequence is: DBAC

Q62. Solution
Correct Answer: (A)

The relationship is that yoga is a type of exercise, just as pineapple is a type of fruit.

Q63. Solution
Correct Answer: (A)

"Elusive" means difficult to find, catch, or achieve. The other options (Elusief, Elusivie, Ellusive) are incorrect spellings.

Q64. Solution
Correct Answer: (B)

The sentence would read: "The convention was called off due to the pandemic as they wanted people to remain safe." "Called off" is a suitable phrasal verb here, meaning the event was canceled.

Q65. Solution
Correct Answer: (A)

The passage states: "Beyond our bilateral relationship, India's more active participation in global economy is needed. India certainly deserves a far bigger share of world trade considering its vast resources. It is strongly hoped that the Indian government's recently initiated effort of enlarging its export market would bear fruit." This clearly supports option (1) because the author encourages India's active participation in world trade, highlighting its vast resources

Q66. Solution
Correct Answer: (D)

The garbage was first sundried for one to three days to bring down the moisture level.

Q67. Solution
Correct Answer: (A)

In Voice change, one can transform the sentence from Active to Passive or vice versa, but can't change the sense. That is, one can't remove the information conveyed through the sentence or add any additional information. That is, transformation must be done keeping the information intact. This is the reason behind selecting (a) as the answer.

Q68. Solution
Correct Answer: (B)

The error is in part (C). The phrase "to do than even the debasing of the coinage than with the financial difficulties" is incorrect because "to do" should be followed by "with" rather than "than." The correct sentence should be: "The fear of an impending invasion has more to do with the debasing of the coinage than with the financial difficulties."

Q69. Solution
Correct Answer: (A)

Every where in the passage we find that the author favours India gaining an edge over China. The author, throughout the passage, is highlighting China's own prospective while they are helping the Africans.

Q70. Solution
Correct Answer: (C)

In this question, you're asked to find the mirror image of the figure when the mirror is placed along the MN line. Analyzing each option: The mirror image will reflect the elements horizontally across the MN line. The circle on top should appear on the right side in the mirror image. The two squares, which are on the right in the original figure, should appear on the left in the mirror image. Comparing this to the options, Option C (labeled as A) accurately represents the mirror image, with the circle on the right side and the squares on the left side. Thus, the correct answer is (3) A.

Q71. Solution

Correct Answer: (C)

The most appropriate word to fill the blank is (3) superior because "intellectually superior" aligns with the description of essays that represent various levels of complexity.

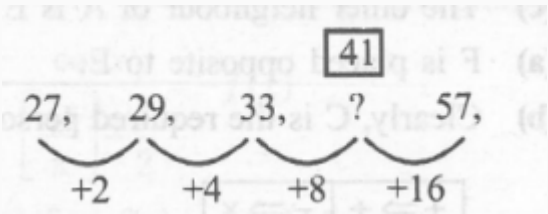
Q72. Solution

Correct Answer: (C)

In this question, you need to identify the "odd one out" among the four options. Analyzing each option: Option A (S2): This combination is symmetrical along the vertical axis. Option B (BB): Both letters "B" are mirror images along the vertical axis. Option C (TL): The combination "TL" does not have symmetry like the others. Option D (LJ): This combination also lacks vertical symmetry. Thus, the odd one out is Option C (TL) because it does not exhibit vertical symmetry as the others do.

Q73. Solution

Correct Answer: (B)



Q74. Solution

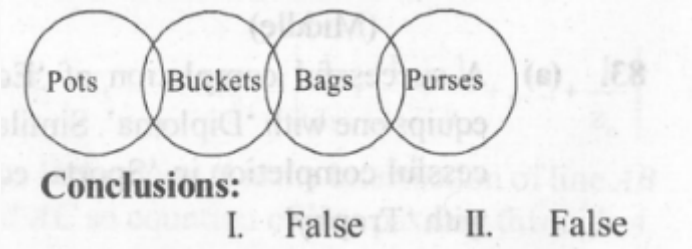
Correct Answer: (D)

The male members in the family are:- (i) The man himself (ii) his four sons; and (iii) his $(3 \times 4) = 12$ grandsons. Hence total number of male members $= 1 + 4 + 12 = 17$

Q75. Solution

Correct Answer: (B)

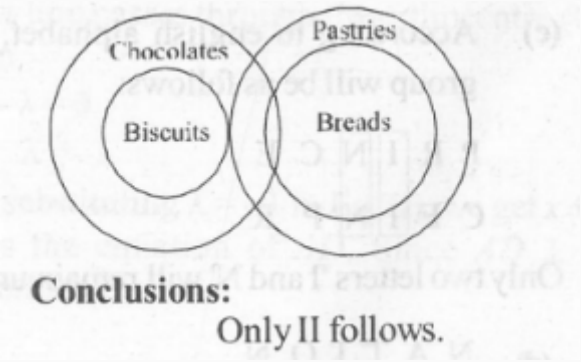
Venn-Diagram Representation:



Q76. Solution

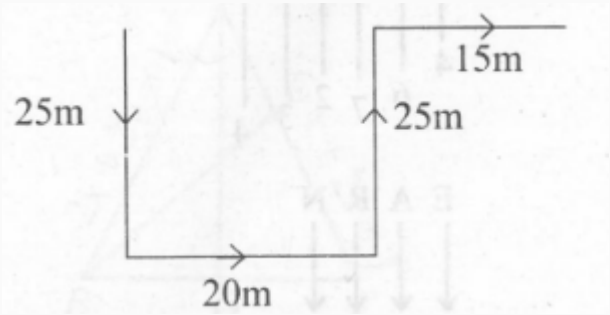
Correct Answer: (C)

Venn-Diagram Representation:



Q77. Solution

Correct Answer: (A)

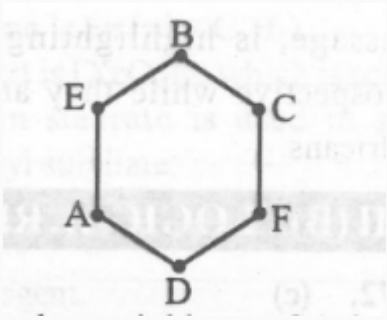


$\therefore$ Rohit is 35 m east.

Q78. Solution

Correct Answer: (C)

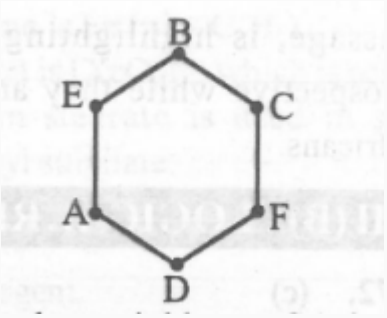
The following hexagonal arrangement is possible.



The other neighbour of A is E.

Q79. Solution

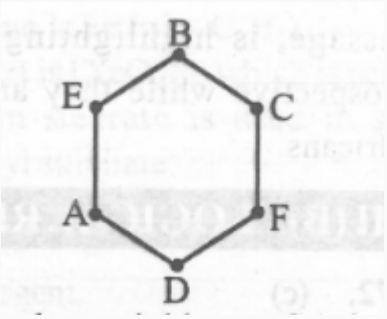
Correct Answer: (B)



The following hexagonal arrangement is possible.
Clearly, C is the required person.

Q80. Solution

Correct Answer: (A)



The following hexagonal arrangement is possible.
F is placed opposite to E.

Q81. Solution

Correct Answer: (B)

$+ \Rightarrow \div$	$- \Rightarrow \times$
$\div \Rightarrow +$	$\times \Rightarrow -$

$63 \times 24 + 8 \div 4 + 2 - 3 =?$ or, $? = 63 - 24 \div 8 + 4 \div 2 \times 3$ or, $? = 63 - 3 + 2 \times 3$ or, $? = 66$

Q82. Solution

Correct Answer: (A)

A successful completion of 'Education' equips one with 'Diploma'. Similary, a successful completion in 'Sports' equips one with 'Trophy'.

Q83. Solution

Correct Answer: (A)

Arrangement as per height: C>D>A>E>B
(Middle)

C is the tallest, A is taller than E but shorter than D, and B is shorter than E.

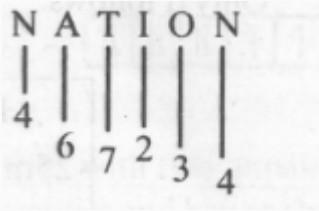
Q84. Solution

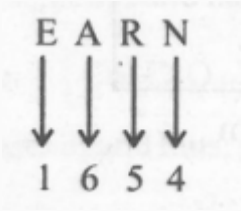
Correct Answer: (C)

From both dices we can see that 4 is opposite to 1,5 is opposite to 3 . Hence correct answer is option (3) 5

Q85. Solution

Correct Answer: (D)





Q86. Solution

Correct Answer: (B)

According to english alphabet, resultant group will be as follows:
Only two letters 'I and N' will remain unchanged.

Q87. Solution

Correct Answer: (B)

$12, 18, 30$ is multiple of 6 . $16, 32, 40$ is multiple of 8 . $36, 18, 27$ is multiple of 9 .

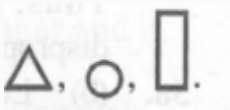
Q88. Solution

Correct Answer: (C)

The series is babb/bbab/bbba/bbbb. Thus, in each sequence, ' a ' moves one step forward and 'b' takes its place and finally in the fourth sequence, it is eliminated.

Q89. Solution

Correct Answer: (C)

Required area common to  .

ie. 9

Q90. Solution

Correct Answer: (A)

The colour of clean sky is blue and blue means green. Hence, the colour of clean sky is green.

Q91. Solution

Correct Answer: (C)

$A=\frac{1}{1-r^a} \Rightarrow 1-r^a=\frac{1}{A} \Rightarrow r^a=1-\frac{1}{A}=\frac{A-1}{A}$ $B=\frac{1}{1-r^b} \Rightarrow 1-r^b=\frac{1}{B} \Rightarrow r^b=1-\frac{1}{B}=\frac{B-1}{B}$
$\therefore a \log r=\log\left(\frac{A-1}{A}\right)$ and $b \log r=\log\left(\frac{B-1}{B}\right) \therefore \frac{\mathrm{a}}{\mathrm{b}}=\frac{\log\left(\frac{\mathrm{A}-1}{\mathrm{A}}\right)}{\log\left(\frac{\mathrm{B}-1}{\mathrm{B}}\right)}=\log_{\frac{\mathrm{B}-1}{\mathrm{B}}}\left(\frac{\mathrm{A}-1}{\mathrm{A}}\right)$

Q92. Solution

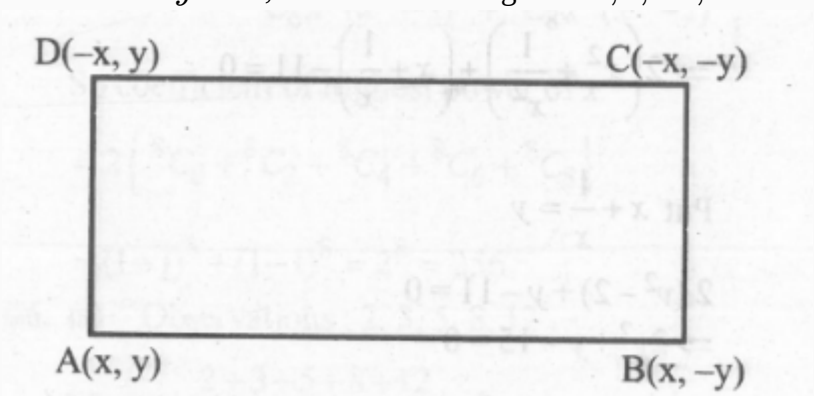
Correct Answer: (C)

Let $S=1+\frac{5}{6}+\frac{12}{6^2}+\frac{22}{6^3}+\frac{35}{6^4}+\ldots$ .....(i) $\frac{S}{6}=\frac{1}{6}+\frac{5}{6^2}+\frac{12}{6^3}+\frac{22}{6^4}+\ldots$ ....(ii) On subtracting (i) from (ii)
$\frac{5}{6} S=1+\frac{4}{6}+\frac{7}{6^2}+\frac{10}{6^3}+\frac{13}{6^4}+\cdots \frac{5}{36} S=\frac{1}{6}+\frac{4}{6^2}+\frac{7}{6^3}+\frac{10}{6^4}+\frac{13}{6^5}+\cdots$ on subtraction
$\frac{25}{36} S=1+\frac{3}{6}+\frac{3}{6^2}+\frac{3}{6^3}+\ldots=\frac{8}{5}$ $S=\frac{8}{5} \times \frac{36}{25}=\frac{288}{125}$

Q93. Solution

Correct Answer: (A)

Let a complex number, $z=x+iy \Rightarrow \bar{z}=\bar{x}-iy$ Then, vertices of rectangle for $z, \bar{z},-z,-\bar{z}$ are



$(x, y),(x,-y),(-x,-y),(-x, y)$
Now, Area of rectangle $=(2x)(2y)=4xy$ It is given that, Area $=2\sqrt{3}=4xy \Rightarrow 2xy=\sqrt{3}$
$\therefore x=\frac{1}{2}, y=\sqrt{3} \quad \therefore z=\frac{1}{2}+\sqrt{3} i$

Q94. Solution

Correct Answer: (D)

$(\sin x+\sin 4x)+(\sin 2x+\sin 3x)=0 \Rightarrow 2\sin\frac{5x}{2}\cdot\cos\frac{3x}{2}+2\sin\frac{5x}{2}\cdot\cos\frac{x}{2}=0$
$\Rightarrow 2\sin\frac{5x}{2}\left\{\cos\frac{3x}{2}+\cos\frac{x}{2}\right\}=0 \Rightarrow 2\sin\frac{5x}{2}\left\{2\cos x\cos\frac{x}{2}\right\}=0$
$2\sin\frac{5x}{2}=0 \Rightarrow \frac{5x}{2}=0,\pi,2\pi,3\pi,4\pi,5\pi,\ldots \Rightarrow x=0,\frac{2\pi}{5},\frac{4\pi}{5},\frac{6\pi}{5},\frac{8\pi}{5},2\pi\ \cos\frac{x}{2}=0 \Rightarrow \frac{x}{2}=\frac{\pi}{2} \Rightarrow x=\pi;$
$\cos x=0 \Rightarrow x=\frac{\pi}{2},\frac{3\pi}{2}$ So, sum $=6\pi+\pi+2\pi=9\pi$

Q95. Solution

Correct Answer: (B)

$\mathrm{p}\rightarrow(\mathrm{q}\vee\mathrm{r})$. If 2 is even number then 2 is a prime number or $2+2=2^2$.

Q96. Solution

Correct Answer: (A)

$$f(-x)=\frac{\cos(-x)}{\left[-\frac{2x}{\pi}\right]+\frac{1}{2}}=\frac{\cos x}{-1-\left[\frac{2x}{\pi}\right]+\frac{1}{2}}$$

(As x is not an integral multiple of π)

$=-\frac{\cos x}{\left[\frac{2x}{\pi}\right]+\frac{1}{2}}=-f(x)$ Therefore, $f(x)$ is an odd function.

Q97. Solution

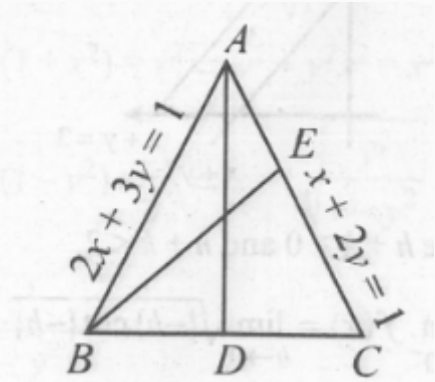
Correct Answer: (C)

$|z_1|=|z_2|=|z_3|=\ldots\ldots=|z_n|=1 \Rightarrow z_1\bar{z}_1=z_2\bar{z}_2=z_3\bar{z}_3=\ldots\ldots=z_n\bar{z}_n=1$
$\bar{\mathrm{z}}_1=\frac{1}{\mathrm{z}},\bar{\mathrm{z}}_2=\frac{1}{\mathrm{z}_2},\bar{\mathrm{z}}_3=\frac{1}{\mathrm{z}_3},\ldots\ldots,\bar{\mathrm{z}}_\mathrm{n}=\frac{1}{\mathrm{z}_\mathrm{n}}$ Now, $|z_1+z_2+z_3+\ldots\ldots+z_n|=|\bar{z}_1+\bar{z}_2+\ldots\ldots+\bar{z}_n|$
$|\bar{z}_1+\bar{z}_2+\bar{z}_3+\ldots.+\bar{z}_n|=\frac{1}{z_1}+\frac{1}{z_2}+\frac{1}{z_3}+\ldots\ldots+\frac{1}{z_n}$

Q98. Solution

Correct Answer: (C)

Here, point A is the intersection of line AB and AC so equation of line passing through A.
$(x+2y-1)+\lambda(2x+3y-1)=0$ .....(i) This line passes through the orthocentre $(0,0)$, then $-1-\lambda=0$
$\Rightarrow \lambda=-1$ On substituting $\lambda=-1$ in Eq. (i), we get $x+y=0$ as the equation of AD. Since $AD \perp BC$,

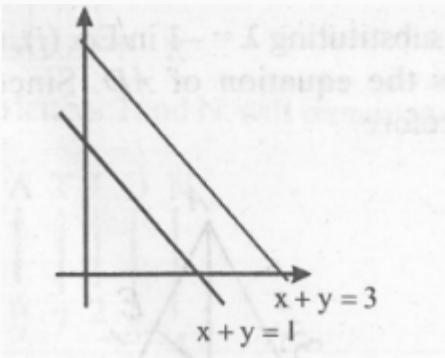


therefore
$-1 \times -\frac{a}{b}=-1 \Rightarrow \quad a+b=0$ ...(ii) Similarly, by applying the condition that BE is perpendicular to CA, we get $a+2b=8$ ....(iii) Now, solving Eqs. (ii) and (iii), we get $a=-8$, $b=8$

Q99. Solution

Correct Answer: (A)

Coordinate of focus will be $(h+1,k)$ Now focus should lie to the opposite side of origin with respect to line

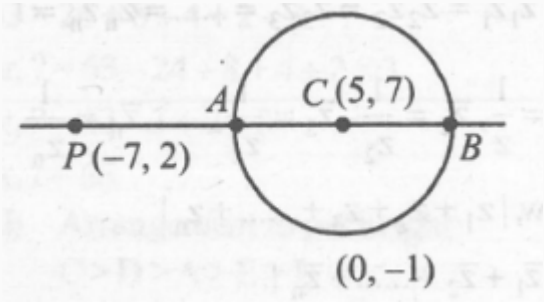


$x+y-1=0$ and same side as origin with respect to line $x+y-3=0$
Hence $h+k>0$ and $h+k<2$.

Q100. Solution

Correct Answer: (A)

The centre C of the circle $=(5,7)$ and the radius $=\sqrt{5^2+7^2+51}=5\sqrt{5}$
PC $=\sqrt{12^2+5^2}=13 \Rightarrow q=PA=13-5\sqrt{5}$ and $p=PB=13+5\sqrt{3}$



$\therefore$ G.M. of p and $q=\sqrt{pq}=\sqrt{(13-5\sqrt{5})(13+5\sqrt{5})}=\sqrt{169-125}=2\sqrt{11}$.

Q101. Solution

Correct Answer: (C)

$^{22}P_{r+1} : {}^{20}P_{r+2} = 11 : 52$ or $(21-r)(20-r)(19-r) = 52 \times 2 \times 21$

$\Rightarrow (21-r)(20-r)(19-r) = 14 \times 13 \times 12$ $\Rightarrow (21-r)(20-r)(19-r) = (21-7)(20-7)(19-7)$

$\Rightarrow r = 7$

Q102. Solution

Correct Answer: (D)

$\lim_{x \to 0^-} f(x) = \lim_{h \to 0} \sqrt{\{-h\} \cot\{-h\}} = \lim_{h \to 0} \sqrt{(1-h) \cot(1-h)} = \sqrt{\cot 1}$

$\lim_{x \to 0^+} f(x) = \lim_{h \to 0} \frac{\tan^2\{h\}}{h^2 - [h]^2} = \lim_{h \to 0} \frac{\tan^2 h}{h^2} = 1 \therefore \lim_{x \to 0} f(x)$ does not exist.

Q103. Solution

Correct Answer: (A)

Given equation can be reduced to a quadratic equation. $\therefore 2x^2 + x - 11 + \frac{1}{x} + \frac{2}{x^2} = 0$

$\Rightarrow 2\left(x^2 + \frac{1}{x^2}\right) + \left(x + \frac{1}{x}\right) - 11 = 0$ Put $x + \frac{1}{x} = y$ $2\left(y^2 - 2\right) + y - 11 = 0 \Rightarrow 2y^2 + y - 15 = 0$

$\Rightarrow y = -3$ and $\frac{5}{2} \Rightarrow x + \frac{1}{x} = -3, x + \frac{1}{x} = \frac{5}{2} \Rightarrow x^2 + 3x + 1 = 0, 2x^2 - 5x + 2 = 0$ Only 2nd equation has rational roots as $\mathrm{D} = 9$ and roots are $\frac{1}{2}$ and 2 .

Q104. Solution

Correct Answer: (A)

Total contested candidates $= 12$ 4 candidates are to be elected and voter votes for at least one candidate then total number of ways of selections. $= {}^{12}\mathrm{C}_1 + {}^{12}\mathrm{C}_2 + {}^{12}\mathrm{C}_3 + {}^{12}\mathrm{C}_4 = 12 + \frac{12 \times 11}{2 \times 1} + \frac{12 \times 11 \times 10}{3 \times 2 \times 1} + \frac{12 \times 11 \times 10 \times 9}{4 \times 3 \times 2 \times 1}$ $= 12 + 66 + 220 + 495 = 793$

Q105. Solution

Correct Answer: (C)

Since $\left(x + \sqrt{x^2 - 1}\right)^8 + \left(x - \sqrt{x^2 - 1}\right)^8 = 2\left\{{}^8C_0 x^8 + {}^8C_2 x^6\left(x^2 - 1\right) + {}^8C_4 x^4\left(x^2 - 1\right)^2 + {}^8C_6 x^2\left(x^2 - 1\right)^3 + {}^8C_8 x^0\left(x^2 - 1\right)^4\right\}$ So coefficient of highest power of $x = 2\left\{{}^8C_0 + {}^8C_2 + {}^8C_4 + {}^8C_6 + {}^8C_8\right\} = (1+1)^8 + (1-1)^8 = 2^8 = 256$

Q106. Solution

Correct Answer: (A)

Observations : $2, 3, 5, 8, 12$ Mean $= \frac{2+3+5+8+12}{5} = 6 \therefore \sigma^2 = 13.2$ Median $= 5 = m \therefore$ Mean deviation about median $\Rightarrow$ $\frac{\Sigma|x_i - m|}{n} = \frac{3+2+0+3+7}{5} = 3$ $M = 3 \Rightarrow \sigma^2 - M = 13.2 - 3 = 10.2$

Q107. Solution

Correct Answer: (D)

Let S denote the set of all triangles in a plane. Let R be the relation on S defined by $(\Delta_1, \Delta_2) \in R \Rightarrow$ triangle $\Delta_1 \cong \Delta_2$: (i) Let any triangle $\Delta \in S$, we have $\Delta_1 \cong \Delta_2 \Rightarrow (\Delta, \Delta) \in R \forall \Delta \in S \Rightarrow R$ is reflexive on (ii) Let $\Delta_1, \Delta_2 \in$ S, such that $(\Delta_1, \Delta_2) \in R$, then $\Delta_1 \cong \Delta_2 \Rightarrow \Delta_2 \cong \Delta_1 \Rightarrow (\Delta_2, \Delta_1) \in R \Rightarrow R$ is symmÄ™tric (iii) Again, let $\Delta_1, \Delta_2, \Delta_3 \in$ S such that $(\Delta_1, \Delta_2) \in R$ and $(\Delta_2, \Delta_3) \in R \therefore \Delta_1 \cong \Delta_2 \cong \Delta_3$
$\therefore (\Delta_1, \Delta_2) \in R \Rightarrow R$ is transitive. $\therefore$ R is an equivalence relation.

Q108. Solution

Correct Answer: (C)

Since $XY = (AB + BA)(AB - BA) = (\mathrm{AB})\mathrm{AB} + (\mathrm{BA})(\mathrm{AB}) - (\mathrm{AB})(\mathrm{BA}) - (\mathrm{BA})(\mathrm{BA})$ Now
$(XY)^{\mathrm{T}} = ((\mathrm{AB}) \cdot (\mathrm{AB}))^{\mathrm{T}} + (\mathrm{BA} \cdot \mathrm{AB})^{\mathrm{T}} - (\mathrm{AB} \cdot \mathrm{BA})^{\mathrm{T}}$
$= (\mathrm{AB})^{\mathrm{T}} \cdot (\mathrm{AB})^{\mathrm{T}} + (\mathrm{AB})^{\mathrm{T}} \cdot (\mathrm{BA})^{\mathrm{T}} - (\mathrm{BA})^{\mathrm{T}}(\mathrm{AB})^{\mathrm{T}}$
$-(\mathrm{BA})^{\mathrm{T}}(\mathrm{BA})^{\mathrm{T}} = \left(B^{\top} \cdot A^{\top}\right)\left(B^{\top} \cdot A^{\top}\right) + \left(B^{\top} \cdot A^{\top}\right) \cdot \left(A^{\top} B^{\top}\right)$
$-\left(\mathrm{A}^{\top}\mathrm{B}^{\top}\right)\left(\mathrm{B}^{\top}\mathrm{A}^{\top}\right) - \left(\mathrm{A}^{\top}\mathrm{B}^{\top}\right)\left(\mathrm{A}^{\top}\mathrm{B}^{\top}\right)$ Since, A & B are symmetric matrix.
$= (\mathrm{BA})(\mathrm{BA}) + (\mathrm{BA})(\mathrm{AB}) - (\mathrm{AB})(\mathrm{BA}) - (\mathrm{AB})(\mathrm{AB}) = (\mathrm{BA} - \mathrm{AB})(\mathrm{BA} + \mathrm{AB}) = -\mathrm{YX}$

Q109. Solution

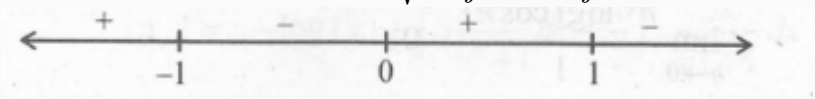
Correct Answer: (C)

Expanding the two determinants, we get $\left(1 - 3x^2 + 2x^3\right) + \left(3x^2 - x^3\right) = 0 \Rightarrow x^3 + 1 = 0$
$\Rightarrow x = -\omega, -\omega^2, -1$ $x^{2007} + x^{-2007} = -1 - 1 = -2$

Q110. Solution

Correct Answer: (C)

Given that, $f(x) = \frac{x}{\sqrt{1+x^2}}$ For injective: Let $x_1, x_2 \in \boldsymbol{R}$ such that $f\left(x_1\right) = f\left(x_2\right)$
$\Rightarrow \frac{x_1}{\sqrt{1+x_1^2}} = \frac{x_2}{\sqrt{1+x_2^2}} \Rightarrow \frac{x_1^2}{1+x_1^2} = \frac{x_2^2}{1+x_2^2} \Rightarrow x_1^2 + x_1^2 x_2^2 = x_2^2 + x_1^2 x_2^2 \Rightarrow x_1^2 = x_2^2 \Rightarrow x_1 = x_2$ So, $f(x)$ is injective. For surjective: Let $y = \frac{x}{\sqrt{1+x^2}} \Rightarrow y^2\left(1 + x^2\right) = x^2 \Rightarrow y^2 + y^2 x^2 = x^2$
$\Rightarrow x^2\left(1 - y^2\right) = y^2 \Rightarrow x = \sqrt{\frac{y^2}{1-y^2}} \Rightarrow \frac{y^2}{1-y^2} \geq 0$



$\therefore y \in (-1, 1)$ So, $f(x)$ is not surjective.

Q111. Solution

Correct Answer: (C)

$\lim_{x \to 0} f(x) = \lim_{x \to 0} \frac{x^2 \log(\cos x)}{\log(1+x)} = \lim_{x \to 0} -\frac{x \cdot \log(\cos x)}{\frac{\log(1+x)}{x}} = \lim_{x \to 0} x \cdot \log(\cos x) = 0 \cdot \log 1 = 0$
$\because \lim_{x \to 0} f(x) = f(0) \therefore f(x)$ is continuous at $x = 0$ Now, $\lim_{x \to 0} \frac{f(x+h) - f(x)}{h}$
$\lim_{h \to 0} \frac{f(h) - f(0)}{h} = \lim_{h \to 0} \frac{h^2 \log \cos h - 0}{\log(1+h)}$ $\lim_{h \to 0} \frac{h^2 \log(\cos h)}{1} = 0 \therefore f(x)$ is differentiable at $x = 0$

Q112. Solution

Correct Answer: (D)

Given $\tan^{-1}\left(\frac{1}{1+1.2}\right)+\tan^{-1}\left(\frac{1}{1+2.3}\right)+\ldots$

$+\tan^{-1}\left(\frac{1}{1+n(n+1)}\right)=\tan^{-1}(x)$

$\Rightarrow \tan^{-1}\left(\frac{2-1}{1+1.2}\right)+\tan^{-1}\left(\frac{3-2}{1+2.3}\right)+\ldots$

$+\tan^{-1}\left(\frac{n+1-n}{1+n(n+1)}\right)=\tan^{-1}(x)$

$\Rightarrow \tan^{-1}(2)-\tan^{-1}(1)+\tan^{-1}(3)-\tan^{-1}(2)+\ldots$

$+\tan^{-1}(\mathrm{n}+1)-\tan^{-1}(\mathrm{n})=\tan^{-1}(x)$ $\Rightarrow \tan^{-1}(\mathrm{n}+1)-\tan^{-1}(1)=\tan^{-1}(x)$

$\left\{\because \tan^{-1}(A)-\tan^{-1}(B)=\tan^{-1}\left(\frac{A-B}{1+A\cdot B}\right)\right\} \Rightarrow \tan^{-1}\left(\frac{n}{n+2}\right)=\tan^{-1}(x) \Rightarrow x=\frac{n}{n+2}$

Q113. Solution

Correct Answer: (B)

Given, $f(x)=5x-3$ and $g(x)=x^2+3$ Let, $y=f(x), \therefore y=5x-3$ $y+3=5x \Rightarrow x=\frac{y+3}{5}$

$\therefore \quad f^{-1}(y)=\frac{y+3}{5} \Rightarrow f^{-1}(x)=\frac{x+3}{5}$ Now, $\quad g(x)=x^2+3$ So, $g \circ f^{-1}(3)=g\left[f^{-1}(3)\right]$

$=g\left(\frac{3+3}{5}\right)=g\left(\frac{6}{5}\right)=\frac{(6)^2}{(5)^2}+3=\frac{36}{25}+3=\frac{111}{25}$

Q114. Solution

Correct Answer: (A)

Given that, $\cot\left(\cos^{-1} x\right)=\sec\left\{\tan^{-1}\left(\frac{a}{\sqrt{b^2-a^2}}\right)\right\}$ Since, $\cos^{-1} x=\cot^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right)$ and

$\tan^{-1} x=\sec^{-1}\left(\sqrt{1+x^2}\right) \Rightarrow \cot\left(\cot^{-1}\frac{x}{\sqrt{1-x^2}}\right)=\sec\left\{\sec^{-1}\sqrt{1+\left(\frac{a}{\sqrt{b^2-a^2}}\right)^2}\right\}$

$\Rightarrow \frac{x}{\sqrt{1-x^2}}=\sqrt{\frac{b^2-a^2+a^2}{b^2-a^2}} \Rightarrow \frac{x}{\sqrt{1-x^2}}=\frac{b}{\sqrt{b^2-a^2}}$ On squaring both the sides, we get $\Rightarrow \frac{x^2}{1-x^2}=\frac{b^2}{b^2-a^2}$

$\Rightarrow x^2b^2-x^2a^2=b^2-b^2x^2$

$x^2b^2+b^2x^2-x^2a^2=b^2 \Rightarrow \quad 2x^2b^2-x^2a^2=b^2 \Rightarrow x^2\left(2b^2-a^2\right)=b^2 \Rightarrow \quad x=\frac{b}{\sqrt{2b^2-a^2}}$

Q115. Solution

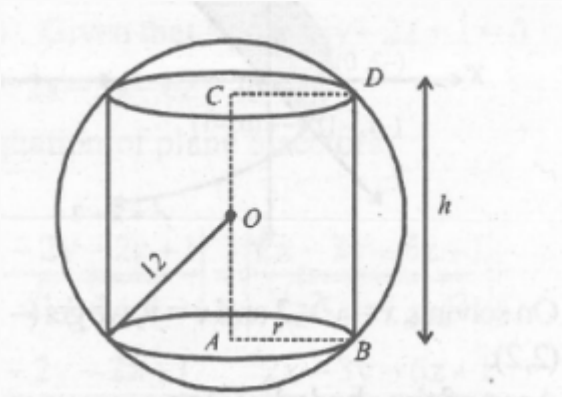
Correct Answer: (A)

$\frac{d}{dx}\left(\tan^{-1}(\cos\sqrt{x})+\sec^{-1}\left(\mathrm{e}^x\right)\right)=\frac{(-\sin\sqrt{x})}{1+\cos^2\sqrt{x}}\cdot\left(\frac{1}{2\sqrt{x}}\right)+\frac{1}{e^x\sqrt{e^{2x}-1}}\cdot e^x$ but when $x=\frac{\pi^2}{4}$

$=-\frac{-\sin\frac{\pi}{2}}{1+\cos^2\frac{\pi}{2}}\left(\frac{1}{2}\right)\left(\frac{2}{\pi}\right)+\frac{1}{\sqrt{e^{\pi^2/2}-1}}=-\frac{1}{\pi}+\frac{1}{\sqrt{e^{\pi^2/2}-1}}$

Q116. Solution

Correct Answer: (B)



$12^2 = r^2 + \left(\frac{h}{2}\right)^2 \Rightarrow V = \pi r^2 h \Rightarrow \quad V = \pi\left(144 - \frac{h^2}{4}\right)h$

$\Rightarrow \quad V = 144\pi h - \frac{\pi}{4}h^3 \Rightarrow \frac{dV}{dh} = 144\pi - \frac{3\pi}{4}h^2 \Rightarrow \frac{dV}{dh} = 0 \Rightarrow 144\pi = \frac{3\pi}{4}h^2$

$\Rightarrow \quad h^2 = 48 \times 4 \Rightarrow h = 8\sqrt{3} \therefore \quad 12^2 = r^2 + 48 \Rightarrow r^2 = 96$ Volume

$= \pi r^2 h = \pi \times 96 \times 8\sqrt{3} = 768\sqrt{3}\pi \text{ cm}^3.$

Q117. Solution

Correct Answer: (B)

Given equations, $x - y + 2z = 4$ $3x + y + 4z = 6$ $x + y + z = 1$ Let $\Delta = \begin{matrix} 1 & -1 & 2 \\ 3 & 1 & 4 \\ 1 & 1 & 1 \end{matrix}$

$= 1(1-4) + 1(3-4) + 2(3-1) = -3 - 1 + 4 = 0$ and $\Delta_1 = \begin{matrix} 4 & -1 & 2 \\ 6 & 1 & 4 \\ 1 & 1 & 1 \end{matrix}$

$= 4(1-4) + 1(6-4) + 2(6-1) = -12 + 2 + 10 = 0$ Now, $\Delta = 0$ and $\Delta_1 = 0 \therefore$ These equations have infinitely many solutions.

Q118. Solution

Correct Answer: (D)

$$I=\int\left(\frac{x^3-1}{x^3+x}\right)dx=\int\left(1-\frac{x+1}{x^3+x}\right)dx \quad \Rightarrow I=\int 1\cdot dx-\int\frac{(x+1)}{x^3+x}dx$$

$$=x-\int\frac{(x+1)}{x\left(x^2+1\right)}dx$$

Let $\dfrac{x+1}{x\left(x^2+1\right)}=\dfrac{A}{x}+\dfrac{Bx+C}{x^2+1}$

$\Rightarrow (x+1)=A\left(x^2+1\right)+(Bx+C)x$ On comparing coefficient, we get

$\Rightarrow (x+1)=(A+B)x^2+Cx+A$

$A+B=0, C=1, A=1$

$\Rightarrow B=-1$

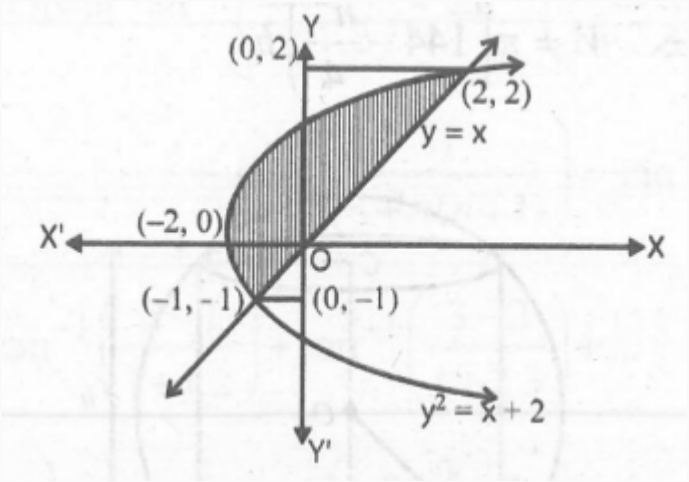
$$\therefore I=x-\int\frac{1}{x}dx-\int\frac{(1-x)}{x^2+1}dx$$

$$\Rightarrow I=x-\log|x|-\int\frac{1}{x^2+1}dx+\frac{1}{2}\int\frac{2x}{x^2+1}dx$$

$$\Rightarrow I=x-\log|x|-\tan^{-1}x+\frac{1}{2}\log\left(x^2+1\right)+c$$

Q119. Solution

Correct Answer: (C)



Given, $x=y^2-2$ and $x=y$.

On solving, $x=y^2-2$ and $x=y$, we get $(-1,-1)$ and $(2,2)$. Area of the shaded region,

$$A=\int_{-1}^{2}ydy-\int_{-1}^{2}\left(y^2-2\right)dy$$

$$=\left[\frac{y^2}{2}-\frac{y^3}{3}+2y\right]_{-1}^{2}=\left(\frac{4}{2}-\frac{8}{3}+4\right)-\left(\frac{1}{2}+\frac{1}{3}-2\right)$$

$$=\frac{10}{3}+\frac{7}{6}=\frac{27}{6}=\frac{9}{2}.$$

Q120. Solution

Correct Answer: (A)

$\frac{dy}{dx} - y\log_e 2 = 2^{\sin x}(\cos x - 1)\log_e 2$ This is linear differential equation

I.F. $= e^{-\log_e 2 \int dx} = e^{-x\log_e 2} = 2^{-x}$ then general Solution is $y2^{-x} = \int 2^{-x}2^{\sin x}(\cos x - 1)\log_e 2dx + c$

$$\therefore y2^{-x} = \log_e 2 \int 2^t dt + c$$

Now let $\sin x - x = t \Rightarrow (\cos x - 1)dx = dt$ $\therefore y2^{-x} = 2^t + c$

$$\therefore y = 2^{x+t} + c2^x$$

$$\therefore y = 2^{\sin x} + c2^x$$

Q121. Solution

Correct Answer: (D)

Here we are given that $\overrightarrow{OA} = \vec{a}, \overrightarrow{OB} = \vec{b}, \overrightarrow{OC} = \vec{c}$ Now P.V of Di.e $\overrightarrow{OD} = \frac{1\times\overrightarrow{OB}+3\times\overrightarrow{OC}}{1+3}$

$$\overrightarrow{OD} = \frac{\vec{b} + 3\vec{c}}{4}$$

$$\overrightarrow{OE} = \frac{4\overrightarrow{OD} + \overrightarrow{OA}}{4+1} = \frac{\frac{4(\vec{b}+3\vec{c})}{4} + \vec{a}}{5}$$

$$\Rightarrow \overrightarrow{OE} = \frac{\vec{a} + \vec{b} + 3\vec{c}}{5}$$

$$\text{Now, } \overrightarrow{OE} = \frac{2\overrightarrow{OB} + 3\overrightarrow{OF}}{2+3}$$

$$\Rightarrow \overrightarrow{OF} = \frac{5\overrightarrow{OE} - 2\overrightarrow{OB}}{3}$$

$$\Rightarrow \overrightarrow{OF} = \frac{\frac{5(\vec{a}+\vec{b}+3\vec{c})}{5} - 2\vec{b}}{3}$$

$$\Rightarrow \overrightarrow{OF} = \frac{\vec{a} - \vec{b} + 3\vec{c}}{3} \Rightarrow P.V. \text{ of F is } \frac{\vec{a} - \vec{b} + 3\vec{c}}{3}$$

Q122. Solution

Correct Answer: (B)

The given equations are $3l + m + 5n = 0$ ...(i) and $6\text{mn} - 2\text{n}l + 5l\text{m} = 0$ ...(ii) From (i), we have $m = -3l - 5n$. Putting $\text{m} = -3l - 5\text{n}$ in (ii), we get $6(-3l - 5n)n - 2nl + 5l(-3l - 5n) = 0$ $\Rightarrow (n + l)(2n + l) = 0 \Rightarrow$ either $l = -\text{n}$ or $l = -2\text{n}$. If $l = -n$, then putting $l = -n$ in (i), we obtain $m = -2n$. If $l = -2\text{n}$, then putting $l = -2\text{n}$ in (i), we obtain $\text{m} = \text{n}$. Thus, the direction ratios of two lines are $-\text{n}, -2\text{n}, \text{n}$ and $-2\text{n}, \text{n}, \text{n}$ i.e., $1, 2, -1$ and $-2, 1, 1$. Hence, the direction cosines are $\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, \frac{-1}{\sqrt{6}}$ or $\frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}$. The angle θ between the lines is given by

$$\cos\theta = \frac{1}{\sqrt{6}} \times \frac{-2}{\sqrt{6}} + \frac{2}{\sqrt{6}} \times \frac{1}{\sqrt{6}} + \frac{-1}{\sqrt{6}} \times \frac{1}{\sqrt{6}} = \frac{-1}{6}$$

$$\Rightarrow \theta = \cos^{-1}\left(\frac{-1}{6}\right)$$

Q123. Solution

Correct Answer: (A)

Let $I=\int_5^9 \frac{\log 3x^2 dx}{\log 3x^2+\log(588-84x+3x^2)}$ ...(i)

$$=\int_5^9 \frac{\log 3x^2 dx}{\log 3x^2+\log 3(14-x)^2}$$

$$=\int_5^9 \frac{\log 3(14-x)^2 dx}{\log 3(14-x)^2+\log 3(14-(14-x))^2}$$

$$\left[\int_a^b f(x)dx=\int_a^b f(a+b-x)dx\right]$$

$I=\int_5^9 \frac{\log 3(14-x)^2 dx}{\log 3(14-x)^2+\log 3x^2}$ ...(ii) Adding Eqs. (i) and (ii), we get

$$2I=\int_5^9 \frac{\log 3x^2+\log 3(14-x)^2}{\log 3(14-x)^2+\log 3x^2}dx$$

$$2\mathrm{I}=\int_5^9 dx=9-5=4\Rightarrow I=2$$

Q124. Solution

Correct Answer: (D)

P (The electronic component fails when first used) $=P(F)=0.10$

$\therefore P(\bar{F})=1-P(F)=0.90$ Let E be the event that a new component will last for one year, then $P(E)=P(F)\cdot P\left(\frac{E}{F}\right)+P(\bar{F})P\left(\frac{E}{\bar{F}}\right)$ [Total probability theorem]

$=0.10\times 0+0.90\times 0.99=0.891$

Q125. Solution

Correct Answer: (B)

Given that $P_1: x-2y-2z+1=0$ $P_2: 2x-3y-6z+1=0$ Equation of plane bisectors

$$\frac{x-2y-2z+1}{\sqrt{1+4+4}}=\frac{2x-3y-6z+1}{\sqrt{2^2+3^2+6^2}}$$

$$\frac{x-2y-2z+1}{3}=\pm\frac{2x-3y-6z+1}{7}$$

Since $\mathrm{a}_1\mathrm{a}_2+\mathrm{b}_1\,\mathrm{b}_2+\mathrm{c}_1\mathrm{c}_2=20>0\therefore$ Negative sign will be taken for acute bisector.

$$\Rightarrow 7x-14y-14z+7=-[6x-9y-18z+3]$$

$$\Rightarrow 13x-23y-32z+10=0$$

$\left(-2,0,-\frac{1}{2}\right)$ satisfy it

Q126. Solution

Correct Answer: (B)

$\hat{i}\times(\bar{a}\times\hat{i})=(\hat{i}\cdot\hat{i})\bar{a}-(\hat{i}\cdot\bar{a})\hat{i}=\hat{j}+2\hat{k}$

Similarly, $\hat{j}\times(\bar{a}\times\hat{j})=2\hat{i}+2\hat{k}$

$\hat{k}\times(\bar{a}\times\hat{k})=2\hat{i}+\hat{j}$

$\Rightarrow \|\hat{j}+2\hat{k}|^2+|2\hat{i}+2\hat{k}|^2+|2\hat{i}+\hat{j}|^2$

$=5+8+5=18$

Q127. Solution

Correct Answer: (C)

(c) We note that $\frac{d}{dx}(x\tan x+1)=x\sec^2 x+\tan x$ $\therefore$ integrating by parts with x^2 as first function, we get

$$I=\int x^2\frac{x\sec^2 x+\tan x}{(x\tan x+1)^2}dx$$
$$=x^2\left(-\frac{1}{x\tan x+1}\right)-\int 2x\left(-\frac{1}{x\tan x+1}\right)dx$$
$$=-\frac{x^2}{x\tan x+1}+2\int\frac{x\cos x}{x\sin x+\cos x}dx$$
$$=-\frac{x^2}{x\tan x+1}+2\log_e|x\sin x+\cos x|+c$$
$$\left(\because\frac{d}{dx}(x\sin x+\cos x)=x\cos x\right)$$

Q128. Solution

Correct Answer: (C)

Given equation of curve $x=12(t+\sin t\cos t), y=12(1+\sin t)^2$ Differentiate w.r.t ' t ',

$$\frac{dx}{dt}=12\left(1+\cos^2 t-\sin^2 t\right)$$
$$\frac{dx}{dt}=12(1+\cos 2t) \text{ and } \frac{dy}{dt}=24(1+\sin t)\cos t$$
$$\frac{dy}{dx}=\frac{2(1+\sin t)\times\cos t}{1+\cos 2t}$$
$$\Rightarrow\frac{2(1+\sin t)\cos t}{2\cos^2 t}=\sqrt{3}$$
$$\Rightarrow t=\frac{\pi}{6}, y_0=27$$

Q129. Solution

Correct Answer: (B)

The statement can be written as: $(A\wedge C)\rightarrow B$ To find the negation of this statement, we use the rule that the negation of an implication $(\mathbf{P}\rightarrow\mathbf{Q})$ is (P $\wedge\neg$Q). Thus, the negation of $(A\wedge C)\rightarrow B$ is: $(A\wedge C)\wedge\neg B$ This means: - "Rishi is a judge" - "Rishi is not arrogant" - "Rishi is not honest" So, the negation of the statement is: "Rishi is a judge, Rishi is not arrogant, and Rishi is not honest."

Q130. Solution

Correct Answer: (D)

For a distribution of random variable x, $\alpha=P\left(X^6<3\right)=P\left(X^6=1\right)+P\left(X^6=2\right)$
$=\lambda+2\lambda=3\lambda$
and $\beta=P\left(X^6<2\right)=P\left(X^6=3\right)+P\left(X^6=4\right)$
$=3\lambda+4\lambda=3\lambda$
$\therefore\quad\alpha:\beta=3:7$